Crowdfunding using Blockchain

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***Abstract*— Crowdfunding has revolutionized the way startups and projects seek funding, while also aiding investors in identifying viable investment opportunities. Through this funding model, small amounts of money can be raised from a large number of investors to reach the desired target. Investors expect a favorable return on their investment, making it a win-win situation for both parties. Online crowdfunding platforms are adept at matching the expectations of fundraisers and investors alike. The growing popularity of blockchain technology in the financial sector offers a decentralized ledger, making transactions more efficient and secure. In order to facilitate transactions using ethers, the crowdfunding platform will utilize the Metamask wallet. Overall, blockchain technology will offer greater transparency and security in transactions.**

***Keywords*— Web3, Metamask, decentralized platform, Blockchain, Solidity, Ethereum, smart contract.**

I. Introduction

In the current era of growing startups, access to capital is critical for their growth. Crowdfunding provides a convenient way for businesses to raise funds without the typical bureaucratic hurdles. Each crowdfunding campaign typically has three essential components: the targeted fundraising amount, a fixed timeframe to raise the funds, and the option for investors to contribute funds to the project. Crowdfunding benefits both the fundraisers and investors, with investors having the opportunity to invest in projects that they believe in, while businesses get the chance to pitch their ideas and raise funds. Crowdfunding involves soliciting donations or investments from individuals for various projects, causes, or companies. Crowdfunding can aid in the growth and advancement of small and medium-sized businesses by enhancing the ease of doing business policies.

Individuals or organizations who want to raise funds for their campaigns or startup projects can present their project ideas, and interested parties can invest in those projects that they believe in. This allows fundraisers to quickly raise the necessary funds without spending time searching for ideal investors interested in their projects. Fundraisers often turn to crowdfunding as a viable alternative to traditional bank loans due to the cumbersome procedures and lengthy processing times involved in obtaining such loans. They seek investors through crowdfunding to raise the necessary funds for their projects. Investors seek promising crowdfunding campaigns or startups with the potential for long-term returns that can benefit them. They typically do not seek financial advisors or intermediaries while evaluating such opportunities.

Although there are advantages to crowdfunding, there are also several drawbacks that need to be addressed. One of the main disadvantages is the potential for fraudulent activities, which can result in investors losing their money. Poor planning by fundraisers can lead to failed projects, which can damage the reputation of both investors and fundraisers and result in losses for both parties. In some cases, intermediaries are involved in the crowdfunding process, which can lead to communication issues and the need to pay

commissions for their services. In crowdfunding events, non-patented projects may be vulnerable to theft of ideas by other organizations. Additionally, in traditional crowdfunding, fundraisers need to personally pitch their ideas to investors, which can be time-consuming and expensive. Fundraisers also need to put a lot of effort into regulating campaigns, marketing, and advertising to find the right investors, which can result in poorly regulated campaigns.

The online crowdfunding platform offers an efficient and transparent solution to overcome many of the limitations of traditional crowdfunding. They allow fundraisers to reach a larger pool of potential investors and provide tools to aid with planning, marketing, and advertising campaigns. These platforms also implement measures to verify the authenticity of projects and protect investors against fraudulent activities. Ultimately, online crowdfunding platforms provide a simpler and more accessible way for both fundraisers and investors to participate in crowdfunding while minimizing risks and maximizing benefits.

II. Background

A. Blockchain technology

Blockchain technology has the potential to transform many industries by providing a decentralized and secure way of recording and verifying transactions. This technology allows for trust and transparency between parties, as each transaction is recorded and verified across a distributed network of computers, rather than relying on a centralized authority or intermediary. The potential applications of blockchain technology are vast, ranging from financial services and supply chain management to healthcare and real estate.

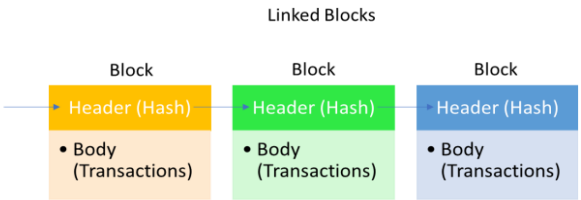

Fig. 1 The figure above shows the simple diagram of how blocks are linked to each other in blockchain

A database typically organizes its information in tables, while a blockchain structures its data into blocks that are linked together. Each block consists of the data of each transaction of digital cash. Each block in the chain contains a unique code, known as a hash, which identifies it and its contents. Once a block is added to the chain, it cannot be modified or deleted without altering all subsequent blocks in the chain. This data structure creates a permanent timeline of information when used in a decentralized context. Once a block is filled with data, it becomes a permanent part of the timeline and cannot be altered.

The architecture of blockchain is based on a decentralized network of nodes, each of which has a copy of the blockchain. The nodes work together to validate new transactions and add them to the blockchain. This process, known as mining, involves solving complex mathematical equations that require significant computational power.

B. Blockchain characteristics

1) *Decentralization*: Decentralization refers to the absence of a centralized authority or intermediary in a blockchain network. In blockchain it refers to the distribution of control and decision-making across a network of participants rather than relying on a central authority. In a decentralized blockchain network, all nodes in the network have a copy of the blockchain ledger and participate in validating transactions and adding new blocks to the chain. This distributed nature of the network provides several benefits, such as greater transparency, increased security, and resilience to attacks. Moreover, it enables peer-to-peer interactions and eliminates the need for intermediaries, resulting in reduced costs and improved efficiency. Decentralization is a crucial feature of blockchain technology, which removes the reliance on powerful central authorities and instead gives control to individual users, making the system fairer and more secure. To ensure that information on the blockchain is consistent and tamper-proof, a set of rules and algorithms, known as consensus protocols, are used for information recording and transaction validation among nodes

in the network. Consensus is achieved when a sufficient number of nodes agree on what information should be recorded on the blockchain. The elimination of a centralized entity responsible for collecting, maintaining, and accessing information improves user privacy and reduces the potential for information misuse. Furthermore, the absence of a central authority for executing and validating transactions results in lower intermediary costs and better performance, as the system operates in a distributed manner, and each node contributes to processing transactions.

2) *Transparency:* One of the most notable features of blockchain technology is its transparency, where all transactions on the ledger are visible to anyone. This transparency is unique to blockchain and provides a high level of accountability and integrity to the information recorded on the ledger. It ensures that no information is unduly altered, deceitfully added, or removed. Such transparency is unprecedented, especially in large financial systems, and fosters trust among the network participants. In crowdfunding, transparency is essential to establish trust between the fundraisers and the investors. Blockchain technology can improve transparency in crowdfunding by creating a decentralized and transparent ledger that records all transactions and activities related to the crowdfunding campaign. This ledger can be viewed by anyone in the network, ensuring that all stakeholders have access to accurate information and can hold each other accountable. This transparency ensures that backers have visibility into how their funds are being used and enables them to monitor the progress of the campaign in real-time. Furthermore, smart contracts can be used to automate the crowdfunding process and ensure that funds are released only when specific milestones are achieved, providing additional transparency and accountability. Overall, the transparency provided by blockchain technology can help to build trust between the fundraisers and the investors, increase transparency and accountability, and reduce the potential for fraud or mismanagement.

3) Security: Security is a key feature of blockchain technology that ensures the integrity and privacy of information stored on the blockchain. The distributed and decentralized nature of the network makes it difficult for attackers to manipulate or corrupt the data on the blockchain. Each node in the network holds a copy of the blockchain, and any changes to the data must be approved by a consensus of nodes in the network, which makes it extremely difficult for malicious actors to tamper with the data. Additionally, the use of cryptographic algorithms and digital signatures in blockchain technology ensures that data stored on the blockchain is secure and private. Transactions on the blockchain are verified using complex mathematical algorithms that ensure that the transaction is valid and the data is not altered. Digital signatures provide an added layer of security, ensuring that only authorized parties can access and modify the data stored on the blockchain. Overall, the combination of decentralization, cryptographic algorithms, and digital signatures makes blockchain technology highly secure and resistant to tampering, making it ideal for applications that require high levels of security and privacy.

C. Smart Contracts In Blockchain

A smart contract is a protocol for computerized transactions that automatically executes the terms of an agreement. The original idea was to convert contractual clauses, such as bonding and collateral, into code and use software or hardware to self-enforce them, reducing the need for trusted intermediaries. Smart contracts are self-executing programs that are stored on a blockchain, such as Ethereum or Hyperledger. They are written in programming languages that are specific to the blockchain platform on which they will run. This allows for a distributed and immutable record of all events, as well as a non-subjective computer code that defines how the process will be managed and what steps will be taken when certain events occur.

One of the main goals of the smart contract in Ethereum was to overcome the limitations of Bitcoin. The smart contract responds to significant events and does not necessarily require the involvement of multiple parties or be legally

binding. It is also referred to as chaincode, and it allows for the programming of rules and decision points into blockchain transactions and processes, automation of transactions, and ensuring that all transactions follow the same rules. When a smart contract is deployed on a blockchain, it becomes a self-executing code that is automatically triggered by predefined conditions. This means that parties involved in a transaction can rely on the code to execute as programmed, without the need for a third party to oversee the process.

This feature of blockchain technology is particularly useful in fundraising scenarios, where creators can create and deploy smart contracts that automatically distribute funds to specific parties based on predefined conditions. This creates trust between the parties involved in the fundraising process, as the rules and conditions for the distribution of funds are clearly defined and enforced by the blockchain.

D. Advantages of using Blockchain for this project

Building a platform on blockchain technology provides numerous benefits. For instance, a crowdfunding platform can:

1) Introduce a blockchain-based voting system that enables shareholders or the crowd to participate in corporate governance in a cost-effective and efficient way.
2) Utilize blockchain-based smart contracts to keep track of all changes in the agreement made between the project initiator and the crowd, thereby allowing regulators to detect fraudulent fundraising.
3) Develop an identity management system that gives users full control over their identity data via blockchain, thus preventing identity theft and money laundering.
4) Implement digital currencies like Bitcoin to eliminate intermediaries such as banks and payment providers, making the payment process quicker and more cost-effective.
5) Establish the conditions under which a transaction takes place, assisting regulators in monitoring and controlling the investment quota and investor qualification.

E. Mathematical concept of EDCSA

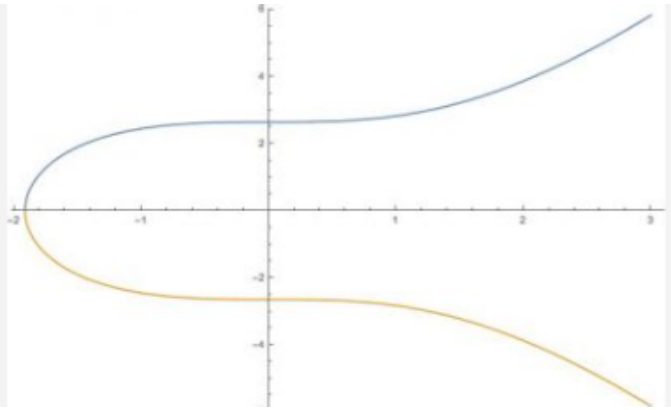

Fig. 2 The figure above shows the Elliptical curve of EDCSA

ECDSA is an acronym for Elliptic Curve Digital Signature Algorithm. It's a cryptographic algorithm used to generate digital signatures for data, such as files, to ensure their authenticity while maintaining their security. In essence, it works like a traditional signature that you can recognize, but it's virtually impossible to counterfeit an ECDSA signature without being detected. Therefore, it provides stronger security guarantees than a physical signature. Blockchain is basically a publicly available ledger where participants enter data and certify their acceptance of the transaction via an elliptic curve digital signature algorithm (ECDSA). An elliptic curve is an equation such as y2 = x3 + a x + b. In Bitcoin and most other implementations, a = 0 and b = 7, so this is simply y2 = x3 + 7 (see graph). Elliptic curves have numerous interesting properties, such as the fact that a nonvertical line intersecting two non tangent points will always intersect a third point on the curve. Indeed, one can define “addition” on the curve as finding that third point corresponding to two given points. This is basically what is done in ECDSA, except that the operations are performed modulo some large prime number M.

In particular, in ECDSA, addition of two points (p1,p2) and (q1,q2), and the doubling of (p1,p2), are performed as follows:
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Addition of (p1,p2) and (q1,q2):
c = (q2 – p2) / (q1 – p1) mod M
r1 = c2 – p1 – q1 mod M

r2 = c (p1 – r1) – p2 mod M

Doubling of (p1,p2):
c = (3 p12) / (2 p2) mod M
r1 = c2 – 2p1
r2 = c (p1 – r1) – p2

Some readers will note that "division" is indicated in the first line of each algorithm. What this means is the product, modulo M, of the expression to the left of the slash by the multiplicative inverse of the expression to the right of the slash. Since M is a prime, every nonzero integer from 1 to M-1 has a multiplicative inverse. For example, the multiplicative inverse of 5 mod 17 is 7, because 5*7 = 35 = 1 mod 17; in other words, 5-1 mod 17 = 7. In practice, these inverses are rapidly calculated by means of the Euclidean algorithm, where one accumulates the divisors in a particular way. See a note by Nick Korevaar for some examples.

One other preliminary detail is how to "multiply" in this algebraic structure, in particular to calculate an expression such as m * (p1,p2) for some integer m. This can be done by first doubling the input (p1,p2), and then using the addition algorithm repeatedly until m copies of (p1,p2) have been added, but this of course is not practical when m and M are very large, as they are in real blockchain applications. Instead, such "multiply" operations are typically done using the binary algorithm for multiplication, which we will sketch here for ordinary integers but which can be easily adapted to ECDSA:

To compute r = n * b mod M: First set t to be the largest power of two such that t is less than or equal to n, and set r = 1. Then perform:

A: If n is greater than or equal to t, set r = b + r mod M, and set n = n – t; else set t = t / 2.

B: If t is greater than or equal to 1 set r = 2 * r mod M, and go to A (if t < 1, then we are done).

Computer scientists will immediately recognize this as almost identical to the binary algorithm for exponentiation modulo M, except that we are adding and doubling instead of multiplying and squaring. It is implied, for example, when one writes 317 mod 10 = ((((32 mod 10)2 mod 10)2 mod 10)2 mod 10) * 3 mod 10 = 3, thus performing the exponentiation in only five multiplications mod 10 instead of 16 or 17.[10]

III. Software architecture

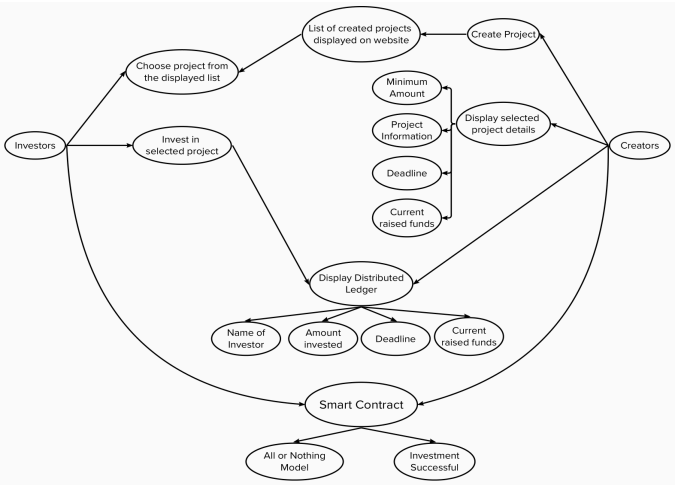

Fig. 3 Use Case Diagram of Crowdfunding using Blockchain

A use case diagram provides an overview of the users, or actors, involved in a system and how they interact with it. In the diagram above, there are two actors: Investors and Creators.

Creators have the ability to create new projects and view information about projects that have been selected by investors. The projects or campaigns created by fundraisers are publicly visible and accessible to everyone. Each campaign includes details such as the fundraising target amount, a description of the campaign with an eye-catching image to attract potential investors, the campaign's deadline for raising funds, and the current amount raised up to the current date. They also have access to the distributed ledger, which is used to track transactions and ensure transparency in the system.

Basic flow of the platform:

1) The first step in creating a campaign is for the creator to connect their Metamask wallet.
2) Next, the creator fills in the project details, such as the name, fundraising goal, deadline, and description.
3) Once the campaign is created, it becomes visible to all potential investors.

4) Interested investors can donate to the campaign they choose using their own Metamask wallet.
5) A smart contract is simultaneously created between the investor and creator on the Web3 dashboard.
6) The Web3 dashboard contains a list of all transactions related to the campaign, making it easy to track donations and monitor progress.

IV. INTEGRATION

Crowdfunding has emerged as a popular method for entrepreneurs and startups to raise funds for their ventures. However, traditional crowdfunding methods have several drawbacks, including high fees and limited access to investors. The emergence of blockchain technology has led to the creation of decentralized crowdfunding platforms that offer increased security, transparency, and accountability. One such platform is the thirdweb, which provides a decentralized web infrastructure that allows developers to build decentralized applications (dApps) that can interact with each other seamlessly.

To integrate crowdfunding with thirdweb, entrepreneurs can create a smart contract that outlines the terms of their crowdfunding campaign, including the fundraising goal, the duration of the campaign, and the rewards for investors. The smart contract is then deployed on the Ethereum network, which is a decentralized platform that enables developers to create smart contracts, decentralized applications, and tokens.

Investors can then contribute to the crowdfunding campaign by sending Ether or other ERC-20 tokens to the smart contract address. The flow of Ether and tokens is managed by the smart contract, which automatically distributes the tokens to investors based on the rewards outlined in the contract. This eliminates the need for intermediaries, such as banks and payment processors, and reduces transaction fees and increases the speed of transactions.

The deployment of smart contracts in the backend on thirdweb using Metamask provides several advantages over traditional crowdfunding methods. First, it enables entrepreneurs to raise funds from a global audience, as investors can participate from anywhere in the world. Second, it provides greater transparency and accountability, as all transactions are recorded on the blockchain and can be audited by anyone. Third, it eliminates the risk of fraudulent fundraising, as the smart contract can keep track of all changes made between the crowd and project initiator.

Integrating the React frontend with the smart contract backend on thirdweb using Metamask is a powerful combination that enables developers to create user-friendly and decentralized crowdfunding platforms. Metamask is a browser extension that provides users with a secure and easy-to-use interface to interact with decentralized applications on the Ethereum network. By combining Metamask with React, developers can create a seamless user experience that allows investors to contribute to crowdfunding campaigns using their Ethereum wallets. The smart contract on thirdweb manages the flow of tokens, while React provides a user interface that displays the relevant information about the crowdfunding campaign and enables investors to track their contributions. This combination of technologies provides a secure, transparent, and efficient way for entrepreneurs to raise funds for their projects, while providing a seamless user experience for investors.

In conclusion, the integration of crowdfunding with thirdweb provides a secure, transparent, and efficient way for entrepreneurs to raise funds for their projects. By leveraging the benefits of blockchain technology and the Ethereum network, crowdfunding platforms can offer a new level of security and transparency that is not possible with traditional crowdfunding methods. The deployment of smart contracts in the backend on thirdweb provides a decentralized, automated way for entrepreneurs to manage the crowdfunding process, from fundraising to token distribution, while reducing the need for intermediaries and increasing the speed of transactions.

V. CONCLUSIONS

The project : Crowdfunding using Blockchain is completed.

Traditional crowdfunding methods have long been plagued by a lack of transparency and fraud. It is a preventable problem, this project has implemented a solid solution that will eliminate these long-standing issues.

The use of blockchain technology is a promising approach to building trust in the crowdfunding process, as it provides a secure and immutable ledger of transactions that can be independently verified. By leveraging the benefits of decentralization and transparency, this project has the potential to increase confidence in the crowdfunding ecosystem and enable more people to participate in supporting good causes.

VI. FUTURE SCOPE

The approach to implement an All-or-Nothing Model as part of the integration in the future. The "All-or-nothing" (AON) crowdfunding model defines that a project can only be initiated if the targeted investment is achieved, otherwise funds will be returned to the funders. In blockchain-based crowdfunding, crypto-currencies can be mined and held by funders until they come across a suitable project. This decentralized system allows fundraisers to pitch projects to currency holders that align with their needs and returns. This model has a higher likelihood of success since projects will either be initiated or not at all. Since the blockchain system is transparent, funders can trust the platform in all aspects, such as project initiation, follow-up returns, and contract clauses.By ensuring that the crowdfunding process is transparent, we are providing donors and investors with clear information about where their funds are going, how they are being used, and the progress of the project.
This transparency can help to build trust between the platform and its users, which is vital for the success of any crowdfunding campaign. Also, on the failure of the campaign the investor is not liable to lose their investment by regaining them in the due time period.

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Ethics Declaration

Conflict of interest

The authors declare they have no conflict of interest.

Ethical Approval

This article does noy contain any studies with human participants or animals performed by any of the authors.

Competing interest

The authors declare no competing interest

Data Availability

No datasets were generated or analysed during the current study